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PROFESSIONAL SUMMARY

Results-driven Software Development Engineer with hands-on experience at Amazon, specializing in building scalable backend systems, LLM-powered pipelines, and real-time personalization engines. Proficient in full-stack development using React.js, Node.js, and AWS cloud services. Strong foundation in Data Structures & Algorithms with proven competitive programming skills. Passionate about designing high-performance, cost-efficient solutions that drive measurable business impact.

PROFESSIONAL EXPERIENCE

Amazon - SDE Intern

Jan 2026 - Present

- Added a runtime-configurable endpoint selector in an internal tool, replacing hardcoded stage-based routing with dynamic backend routing — enabling instant environment switching without redeployment and reducing configuration time from minutes to seconds.
- Building a scalable text generation pipeline using LLMs (Amazon Bedrock) to generate personalized marketing templates for a 10M+ content catalog.
- Reduced inference cost by 89% through model benchmarking across 8 LLM models and implementing a batch processing architecture using AWS Fargate and AWS Step Functions.
- Engineered a real-time template selection engine that matches pre-computed templates against user signals and content metadata at request time, serving personalized strings in <5ms with zero runtime LLM cost.
- Building a configurable priority waterfall per content type, ranking continuity, editorial, and occasion signals to achieve 100% detail page coverage with highest-value messaging.
- Executing an A/B experiment strategy measuring detail-page-to-play conversion across treatments (personalized vs. generic) with CSC and PAC guardrails.

EDUCATION

Chennai Institute of Technology | B.Tech — Information Technology - 8.78 | 2022 - 2026

Coursework: Data Structures & Algorithms, Software Development, Operating Systems, Database Management Systems

PROJECTS

MacroMate | React.js, Express.js, Node.js, MySQL

- Developed a dynamic web application to calculate macronutrients in recipes and track intake using React.js.
- Built ingredient input features with real-time macronutrient breakdown calculations per recipe.
- Designed a recipe management system to add and access favorite recipes, enhancing overall user engagement.
- Implemented a responsive design to ensure a seamless user experience across devices, improving accessibility.

Hostel management System | React.js, REST APIs, TailwindCSS, Express.js, MySQL

- Developed a full-stack hostel management app to automate outpass requests with real time status tracker.
- Implemented RBAC for students, advisors, HODs, and wardens to streamline multi-level approval workflows.
- Built authentication, REST APIs, and routing for seamless frontend-backend communication.
- Designed a responsive UI using TailwindCSS, ensuring optimized experience across desktop and mobile platforms.

TECHNICAL SKILLS

Languages: Java, JavaScript, SQL, HTML5, CSS3

Frameworks & Libraries: React.js, Node.js, Express.js, TailwindCSS

Databases: MySQL, Relational Database Design

Cloud & DevOps: AWS (Bedrock, Fargate, Step Functions, S3), Docker, CI/CD

Tools: Git/GitHub, IntelliJ IDEA, VS Code, Postman

Concepts: Data Structures & Algorithms, REST APIs, Agile/Scrum, OOP

ACHIEVEMENTS & COMPETITIVE PROGRAMMING

- LeetCode Rating: 1706 | Top 13.5% globally | 500+ problems solved across multiple data structure and algorithm topics.
- CodeChef Max Rating: 1410+ | 2-Star | Gained 100+ rating points in a single contest.
- Shortlisted for pre-final round of Techgrium 8th Edition for developing an innovative solution for chronic insomnia.
- Secured Global Rank 2,099 in LeetCode's 431st Weekly Contest (Top 14.86%).